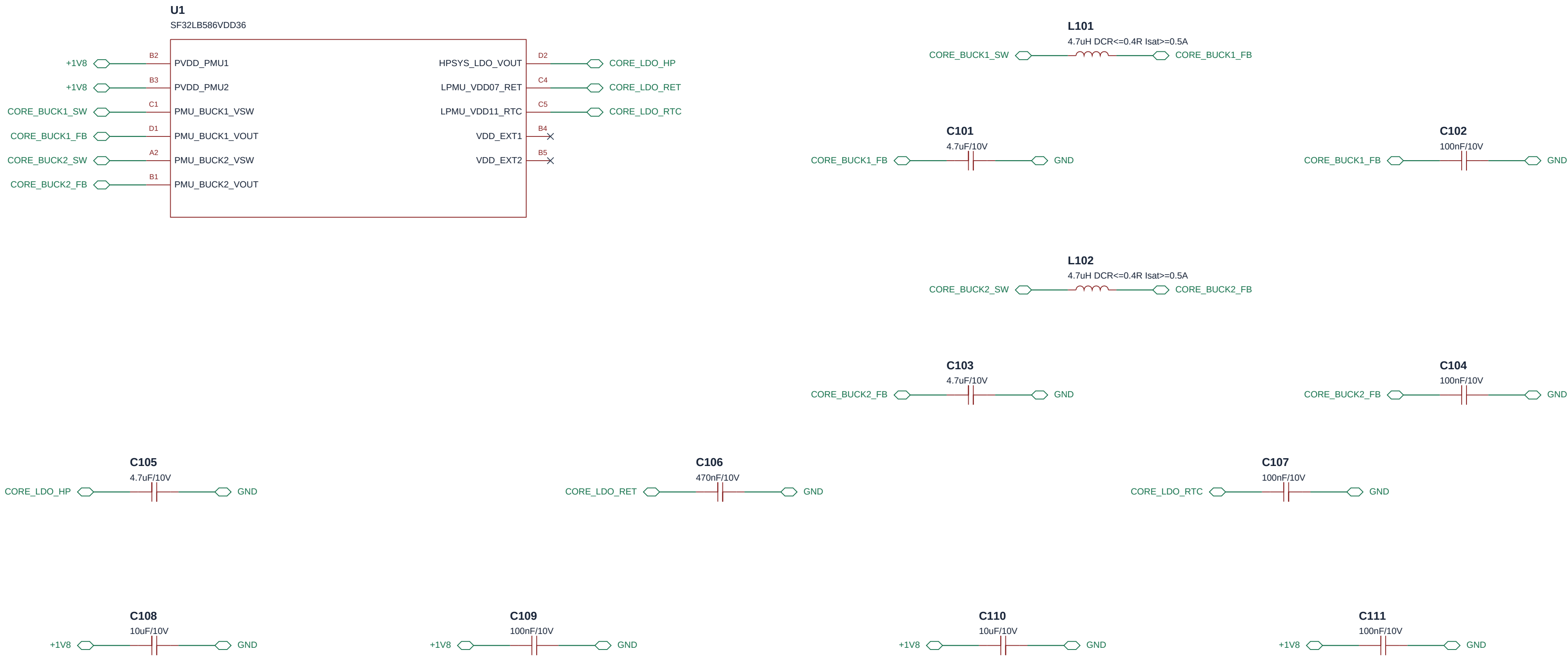


HealthWatch-SF58-BareChip-v0.3 | 01 BARE CHIP PMU

U1 = SF32LB586VDD36 BGA256 | internal buck/LDO support | never apply battery voltage to the 1.8 V rails



L101/C101/C102: BUCK1 VSW -> inductor -> FB. L102/C103/C104: BUCK2 VSW -> inductor -> FB.

The internal LDO nodes only take bypass capacitors. Do not connect them to +1V8 or +3V3.

B4/B5 VDD_EXT remain open as in the official internal-PMU reference. Recheck for any external-core-supply mode.

All 44 ground/reference-return balls are listed on the final sheet; connect to the common ground plane.

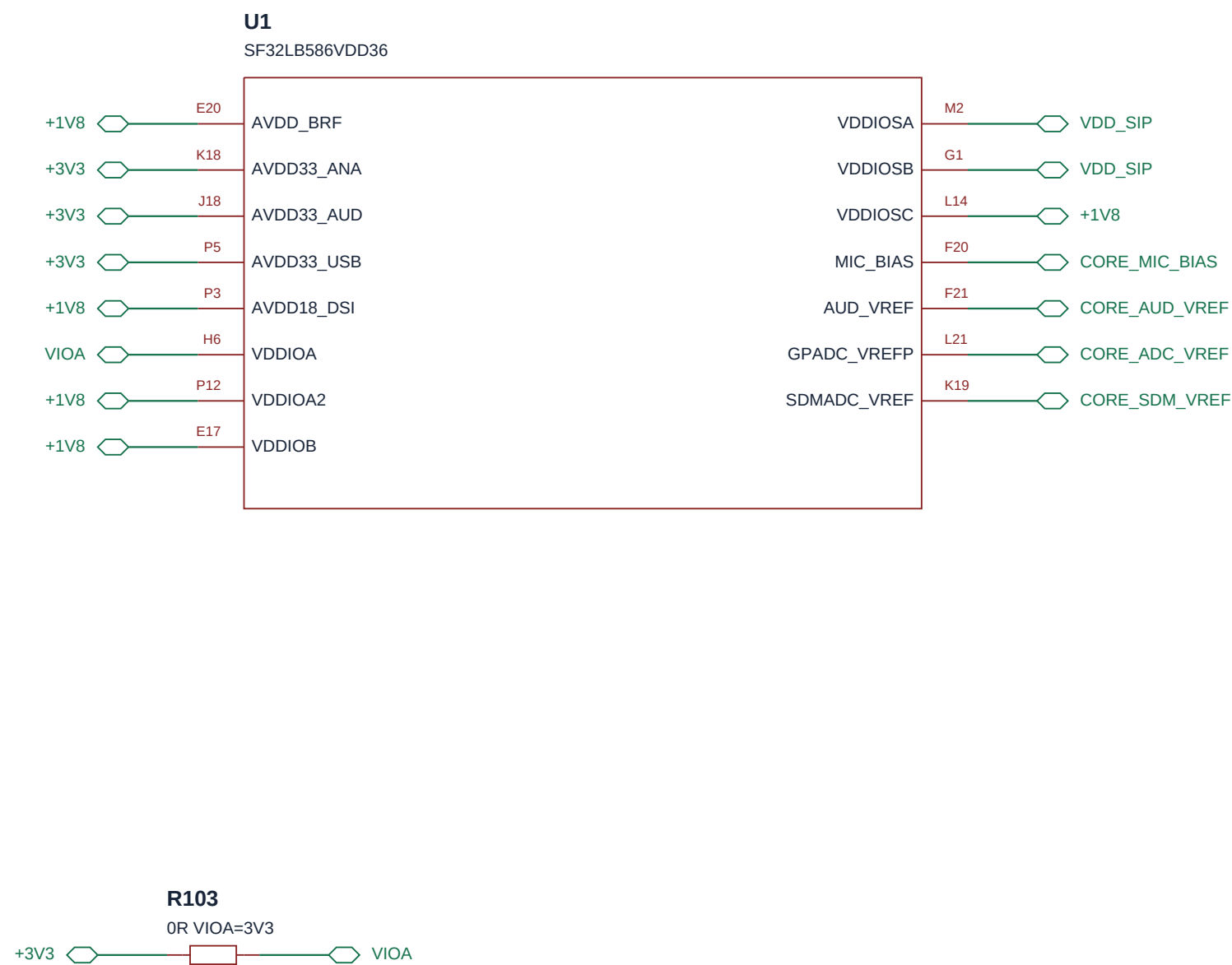
Input power comes from the POWER RAILS sheet. Clock/reset and SiP power switching are shown on subsequent sheets.

BGA footprint is bound; stackup, fanout, crystal layout and assembly process still require PCB review.

U1 is a purchased packaged IC soldered directly onto the custom mainboard; there is no core module in this BOM.

HealthWatch-SF58-BareChip-v0.3 | 02 CHIP POWER DOMAINS

PA12-93 bank fixed at 3.3 V for the selected NOR + kit display adapter | PB / PA0-11 = 1.8 V



VDDIOSA/B = switched 1.8 V for the two SiP PSRAMs; VDDIOSC = always-on 1.8 V for SiP NOR.

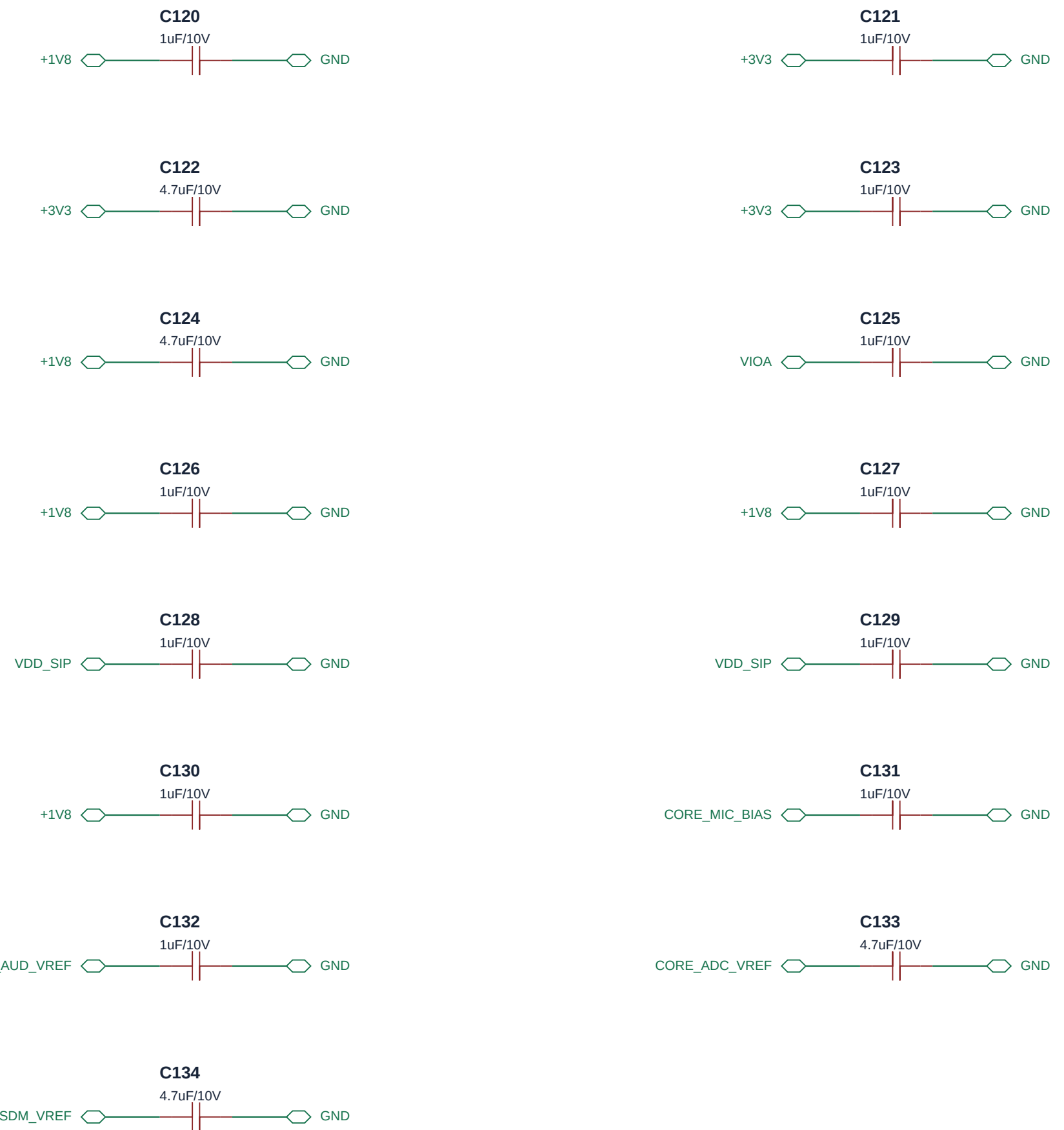
VIOA feeds both the external NOR bus and display/control PA12-93 signals. Do not change to 1.8 V with PY25Q128HA.

A future 1.8 V-only panel requires level translation or a compatible Flash/bank remap; its exact FPC is still pending.

USB/audio/DSI power bypass follows the reference, although those functions are disabled in this offline revision.

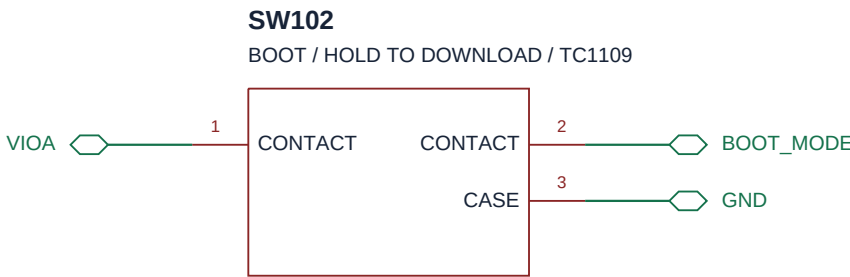
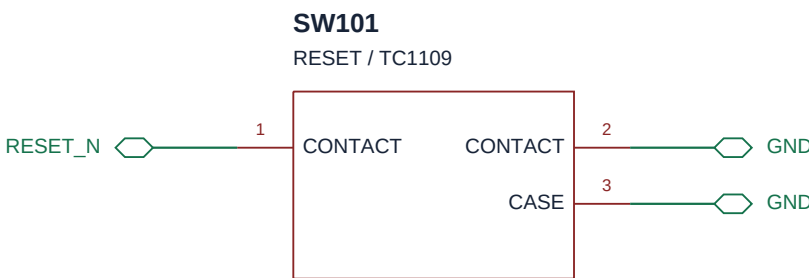
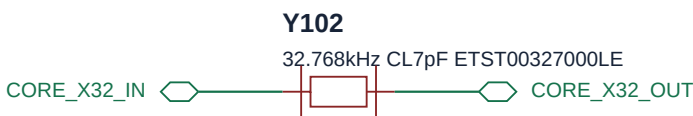
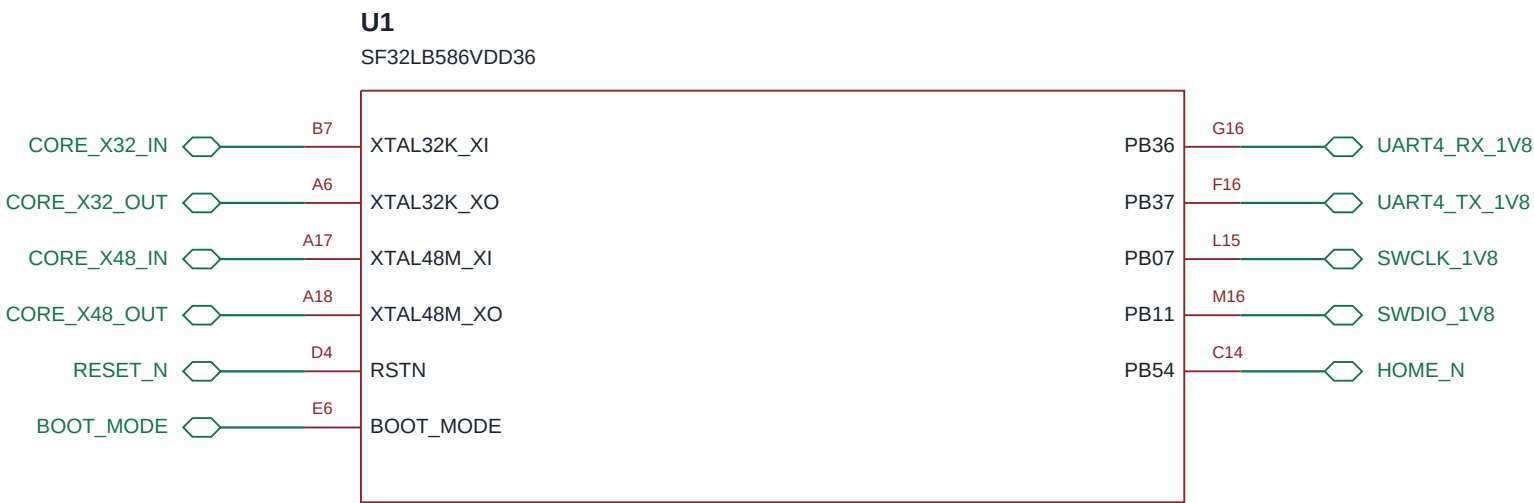
Each capacitor has an associated BGA ball in its engineering note; place locally, not as one remote bulk capacitor bank.

GPADC/AUD/SDM reference nodes are capacitor-only; they are not external power inputs.



HealthWatch-SF58-BareChip-v0.3 | 03 CLOCK RESET AND DEBUG

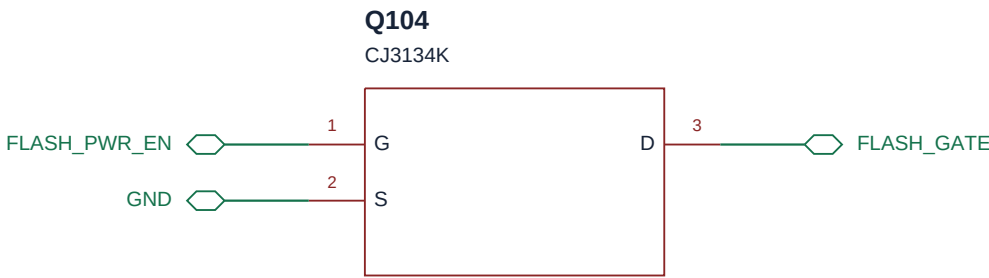
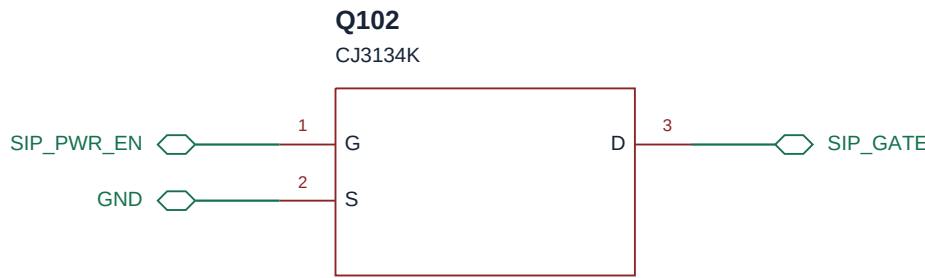
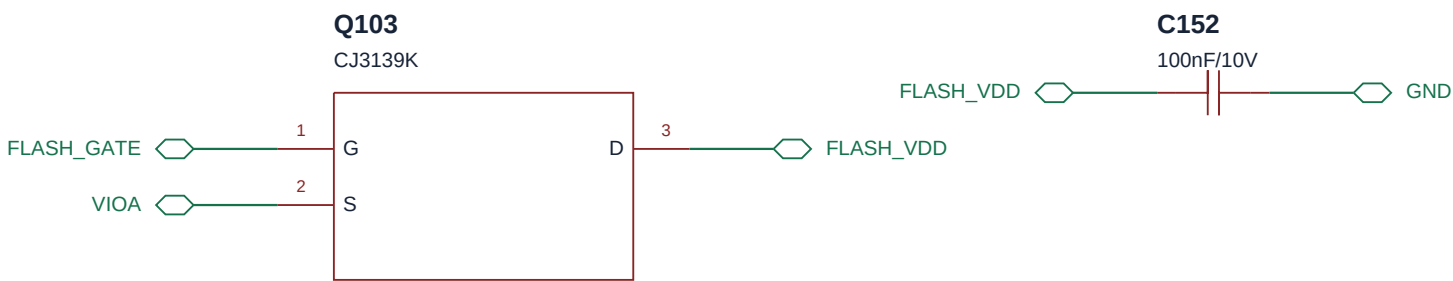
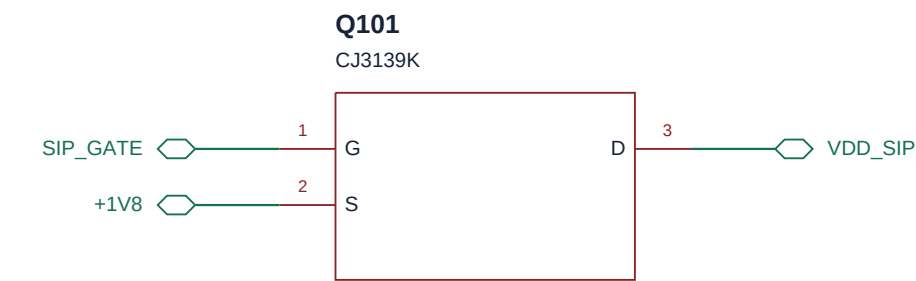
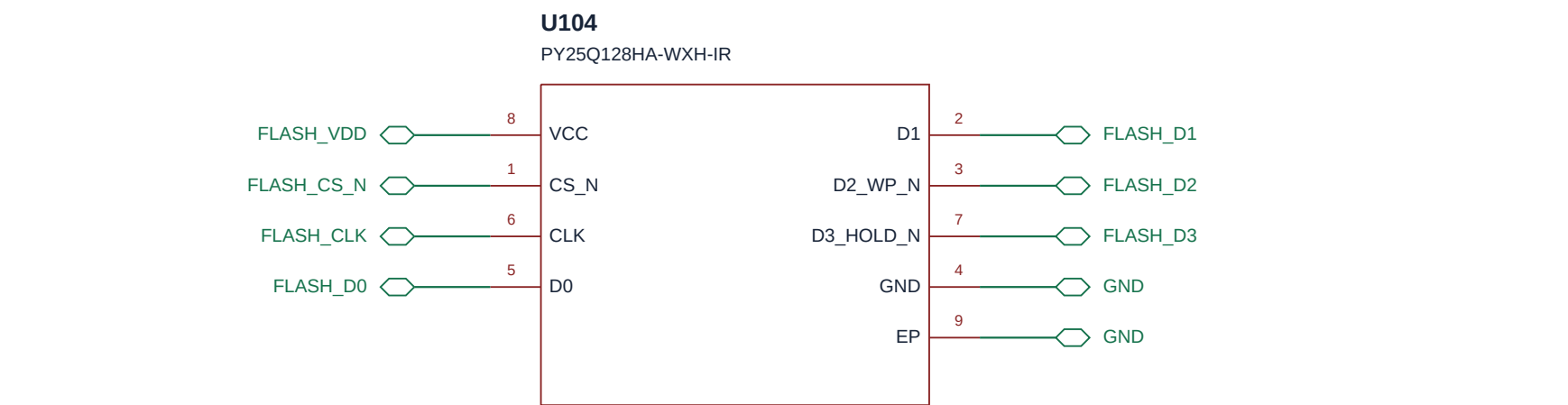
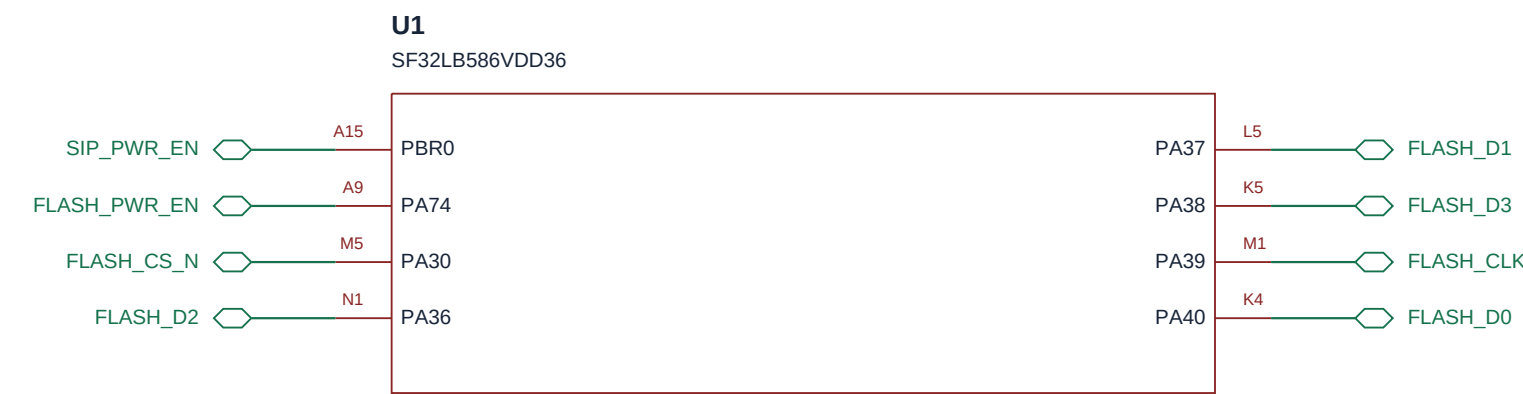
48 MHz + 32.768 kHz crystals | 1.8 V SWD/UART | BOOT_MODE at VIOA = 3.3 V



J101 is for a true 1.8 V debugger/UART adapter. VTREF is a reference output, not an input supply.
RESET_N must be pulled up to PVDD1 (+1V8). BOOT_MODE belongs to VIOA (+3V3).
BOOT low = normal boot, high at reset = download. HOME/PB54 is on DISPLAY AND CONTROLS.
Start with external crystal trim capacitors unpopulated for the selected CL values, then measure/calibrate.
HCPU target up to 240 MHz; first bring-up uses conservative clock and Flash timing.
Budget: 32 MB SiP PSRAM total, 1 MB SiP NOR, plus the external 16 MB NOR on the next sheet.

HealthWatch-SF58-BareChip-v0.3 | 04 SIP AND RESOURCE FLASH

SF32LB586VDD36 contains 16+16 MB PSRAM + 1 MB NOR | U104 adds 16 MB 3.3 V QSPI NOR



SIP PSRAM bus wiring is inside the package. PBR0 controls its 1.8 V switch; the 1 MB SiP NOR stays powered via VDDIOSC.

PA74 controls external Flash power. MPi4 = PA30 CS, PA39 CLK, PA40 D0, PA37 D1, PA36 D2, PA38 D3.

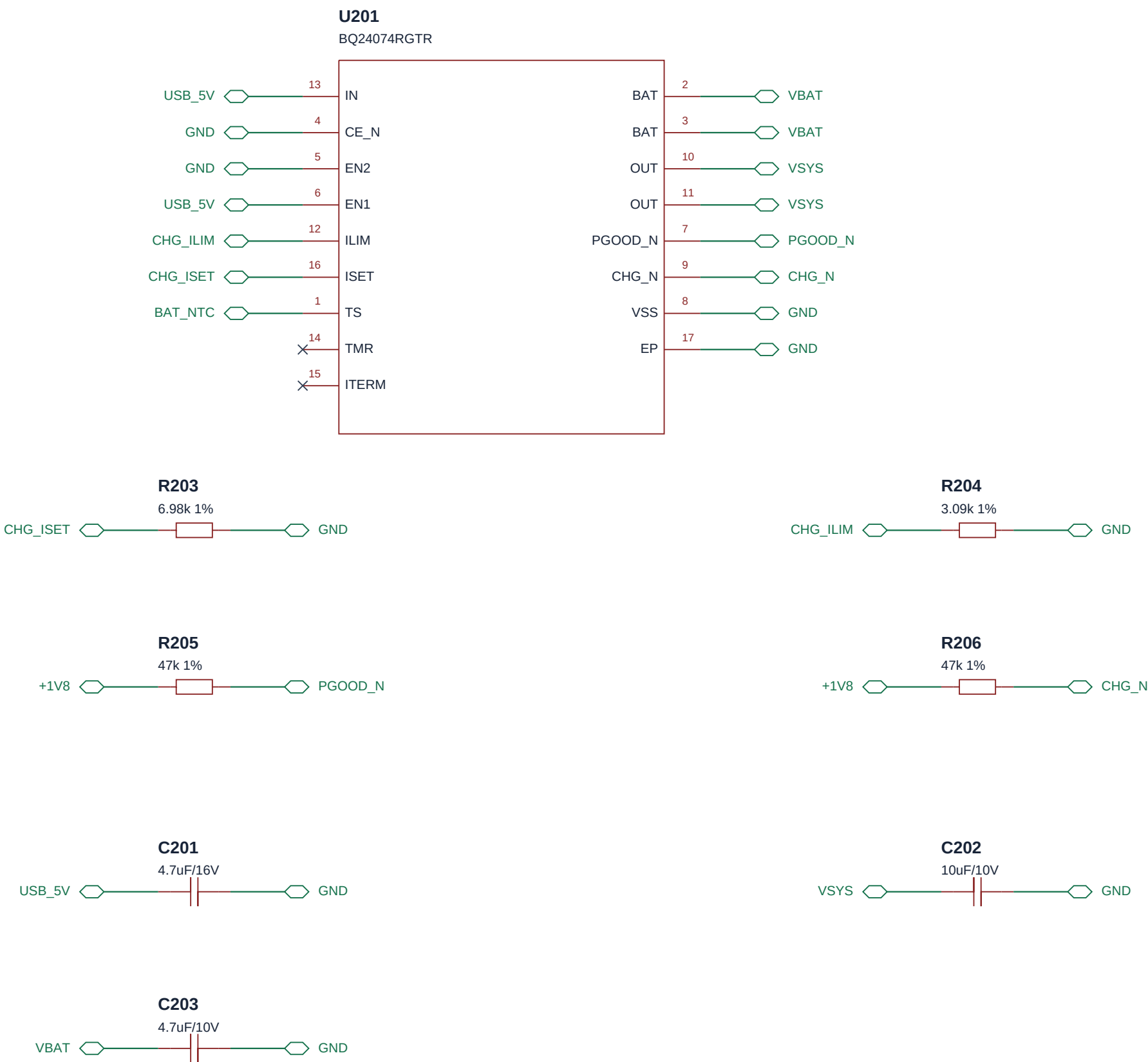
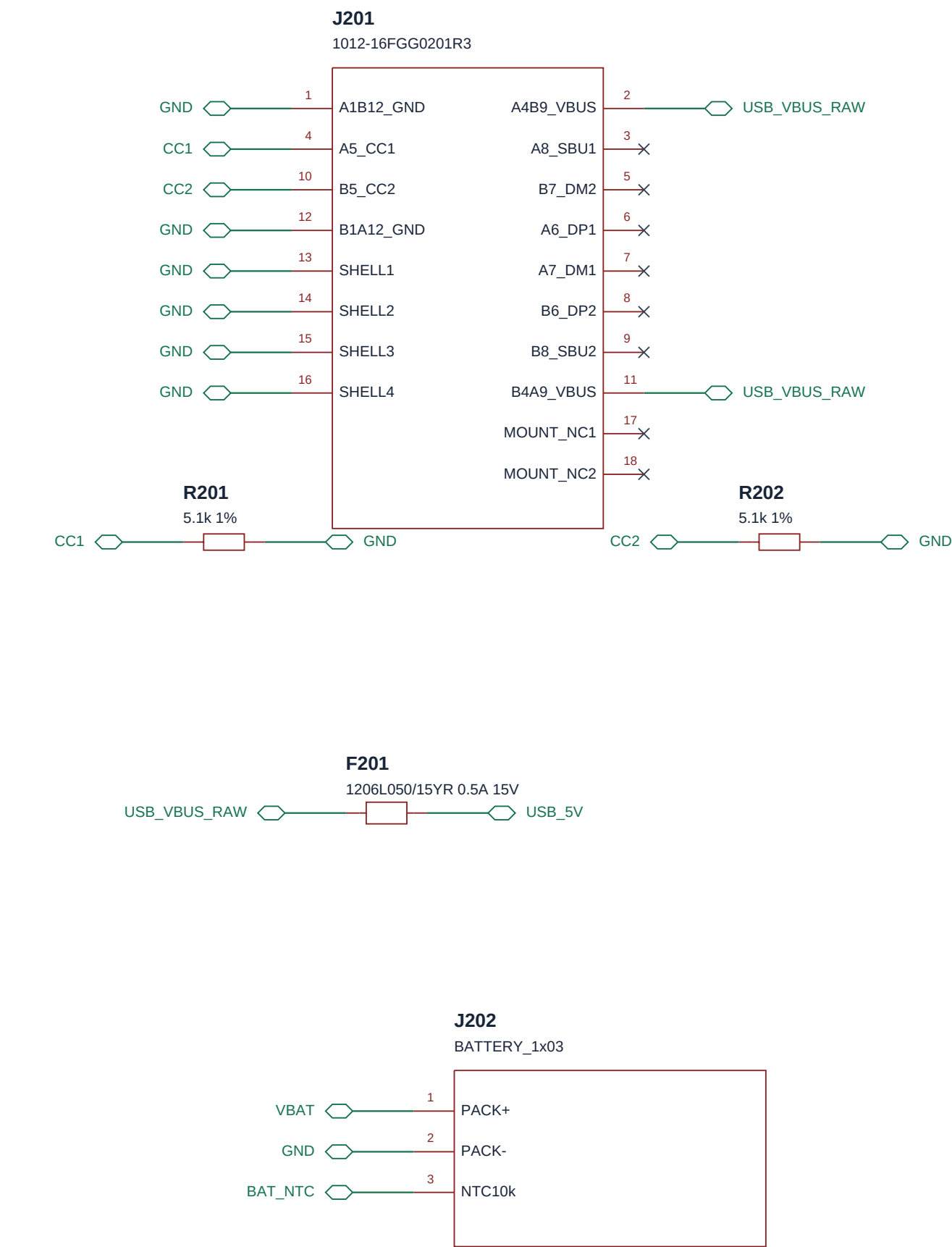
This bank is fixed at 3.3 V. Enable and settle FLASH_VDD before access; place bus pins Hi-Z before power-off.

Pin map/power switch follows the SiFi N16R32N1 reference. Firmware BSP must retain correct SIP and Flash startup control.

Module BOM removed. The final board purchases U1, U104, crystals and passives individually; no 24x24 mm core module.

HealthWatch-SF58-BareChip-v0.3 | 05 BATTERY AND CHARGER

BQ24074 power path | single-cell 4.2 V charge termination | nominal 128 mA initial charge current

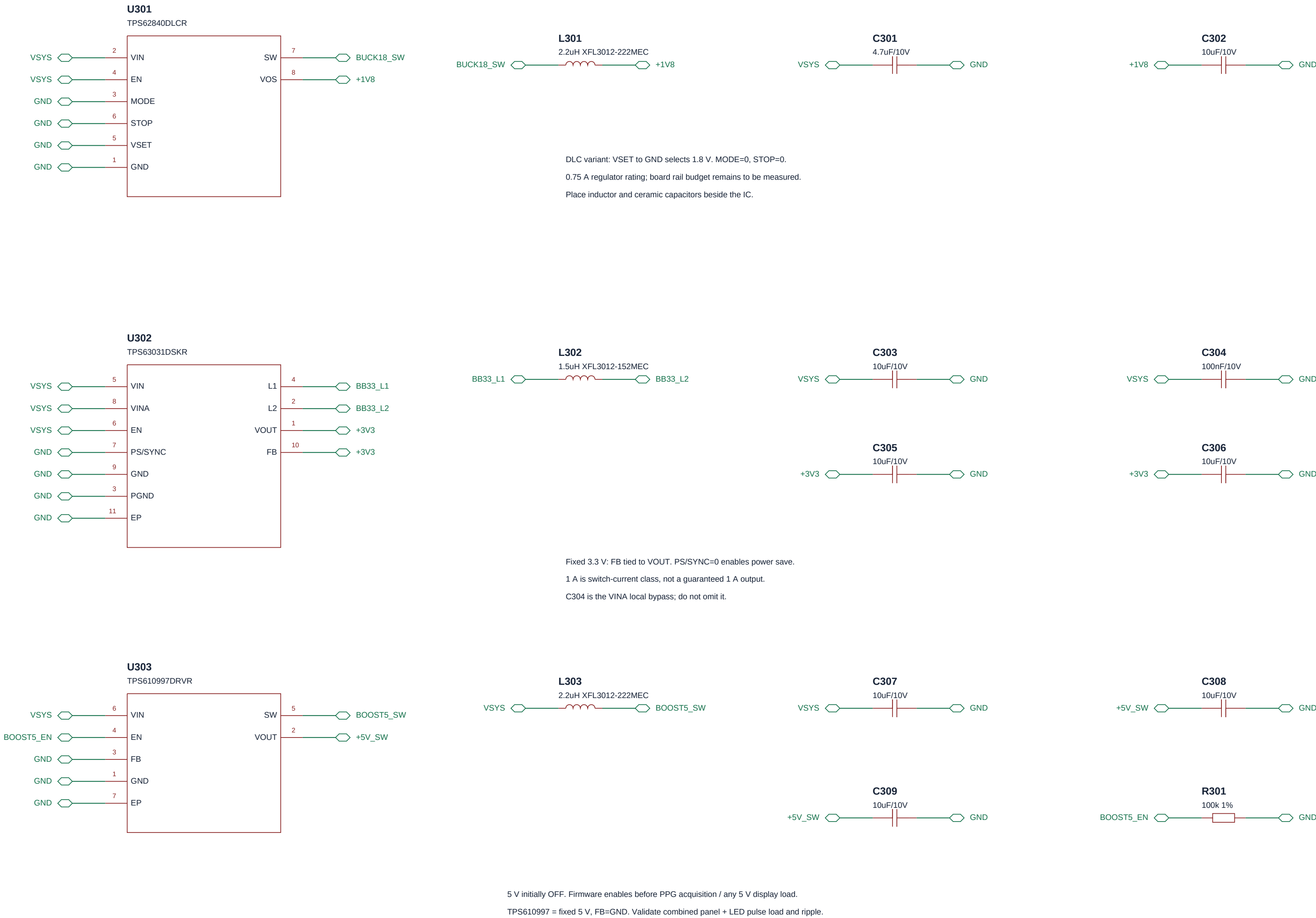


Battery: protected 1S Li-ion/LiPo, 4.2 V charge rating; nominal 3.7 V.
400 mAh is a planning assumption. Verify max charge current and cell dimensions.
Use a real 10k NTC in thermal contact with the cell; TS is not tied to GND.
Connector polarity and NTC B value must be checked against the selected pack.
USB-C is charge-only. USB data / SBU contacts are not routed in this revision.

$I_CHG\ typ = 890 / 6980 = 127.5\ mA$ (charger / resistor tolerance applies).
EN2=0, EN1=1: fixed 500 mA input mode; use a suitable USB 5 V supply.
ILIM resistor fitted as required even when fixed input mode is selected.
TMR and ITERM open: internal default timer / termination behavior.
VSYS ~ battery voltage on battery; up to 4.4 V with external input.
Protection IC is required in the purchased battery pack; not duplicated here.

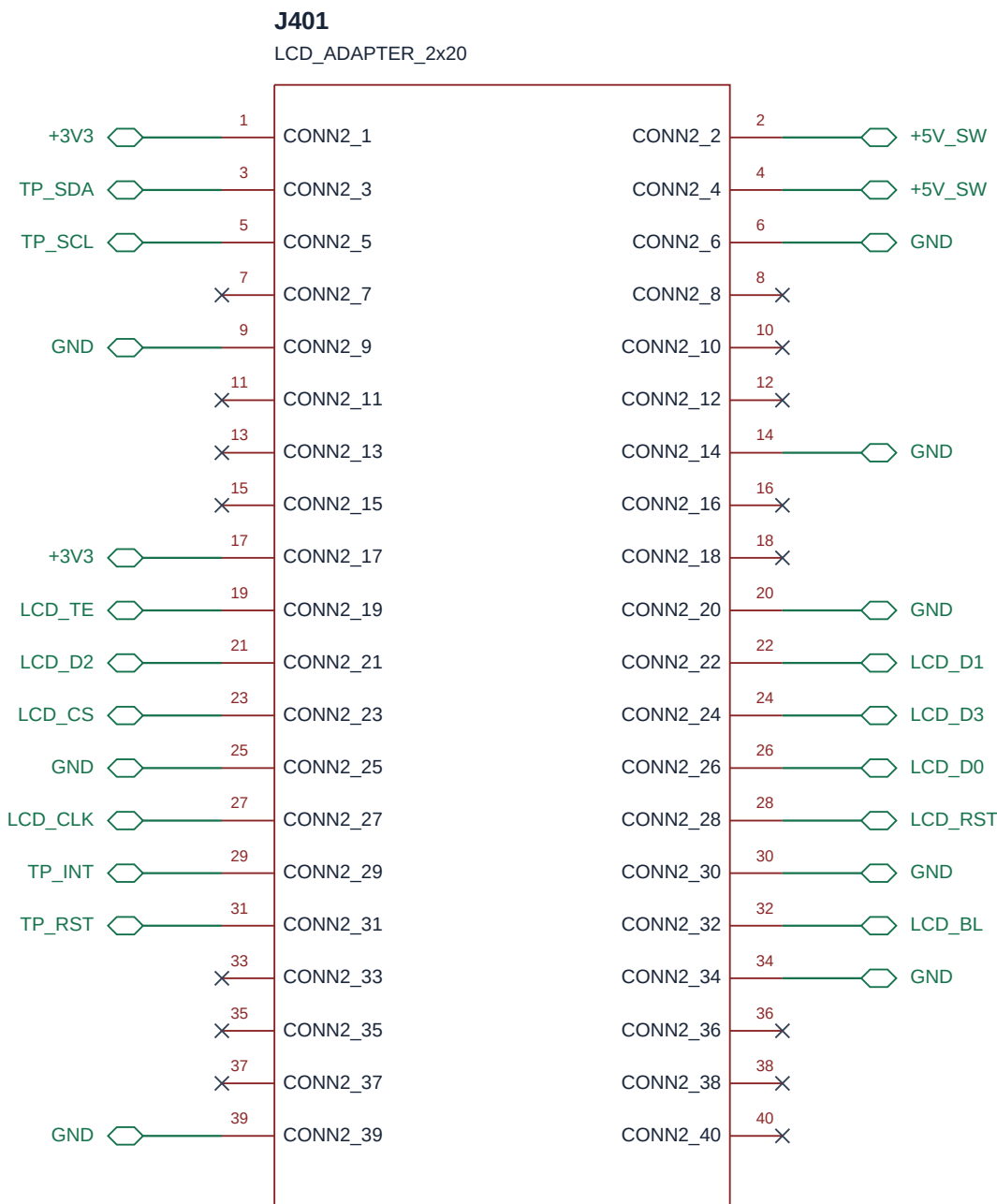
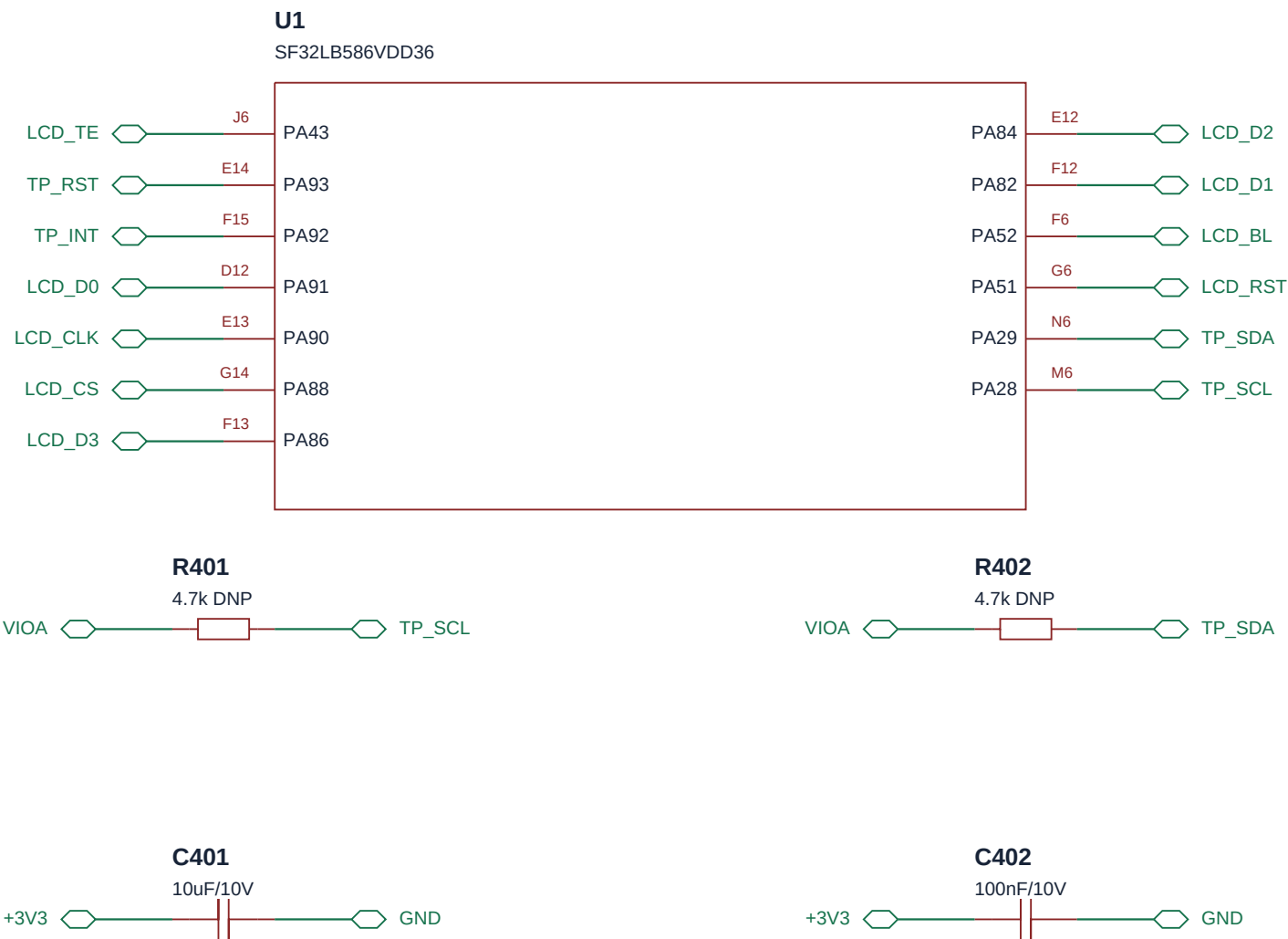
HealthWatch-SF58-BareChip-v0.3 | 06 POWER RAILS

VSYS -> 1.8 V buck, 3.3 V buck-boost, switchable 5 V boost | validate startup and peak current



HealthWatch-SF58-BareChip-v0.3 | 07 DISPLAY AND CONTROLS

390x450 QSPI AMOLED + I2C touch | connector matches required SiFi CONN2 signals; panel FPC is not guessed

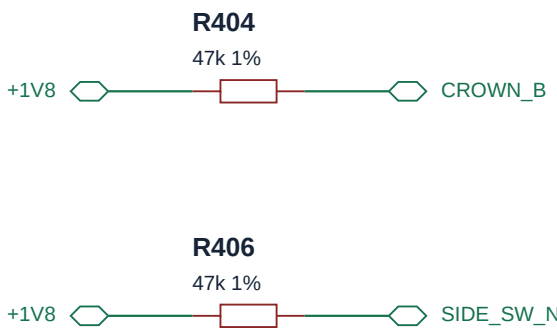
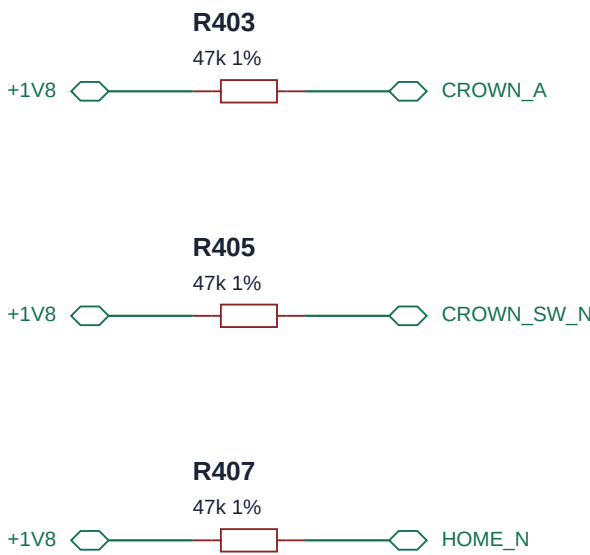
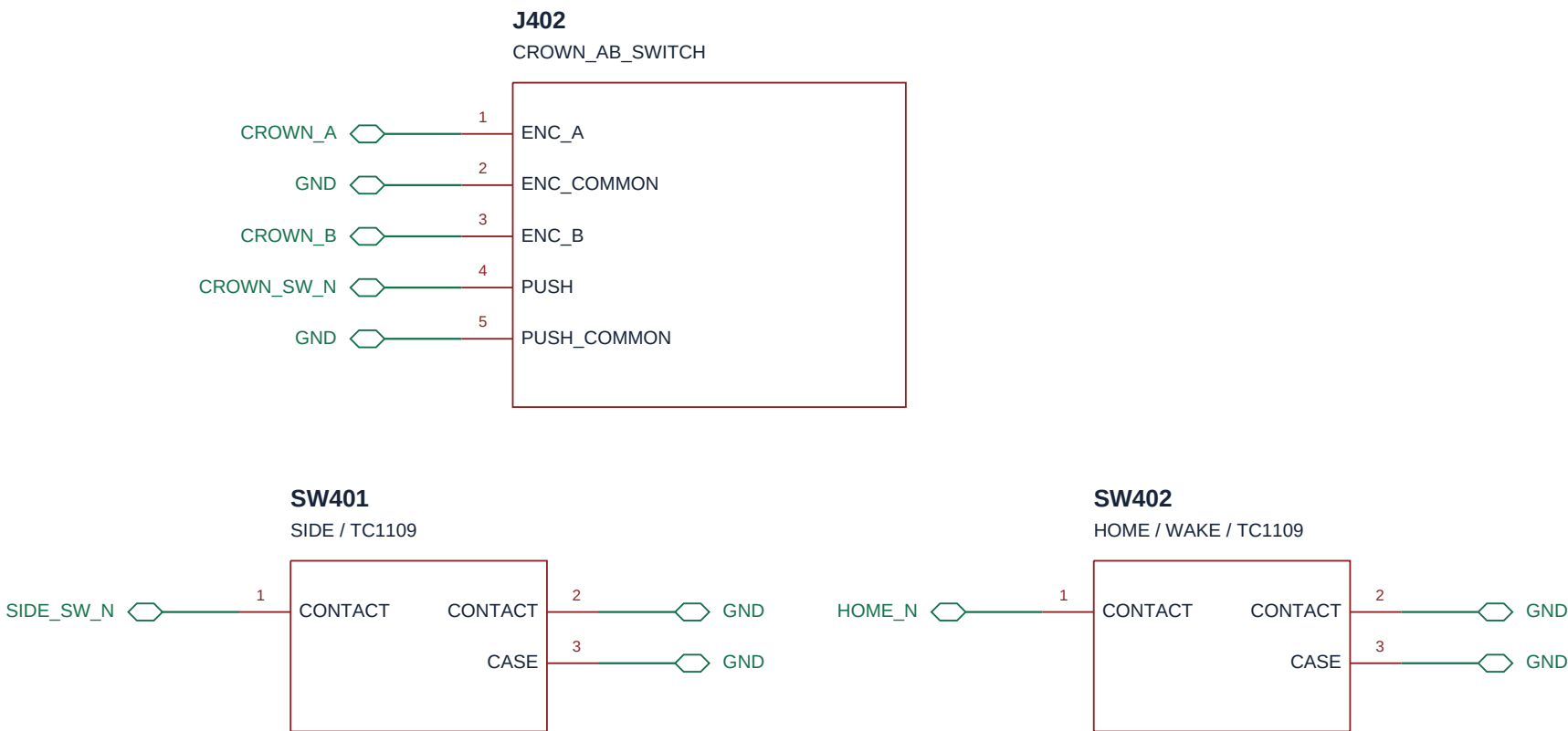


Purchased 1.85in QSPI AMOLED kit confirmed. Use its adapter; check orientation before powering.

5 V shares the PPG boost. Unused CONN2 audio / UART / SWD pins are open.

For naked AMOLED: need its FPC drawing, rails and power-up timing first.

CO5300AF-01 / FT6146-M00 are screenshot data, not a pinout specification.

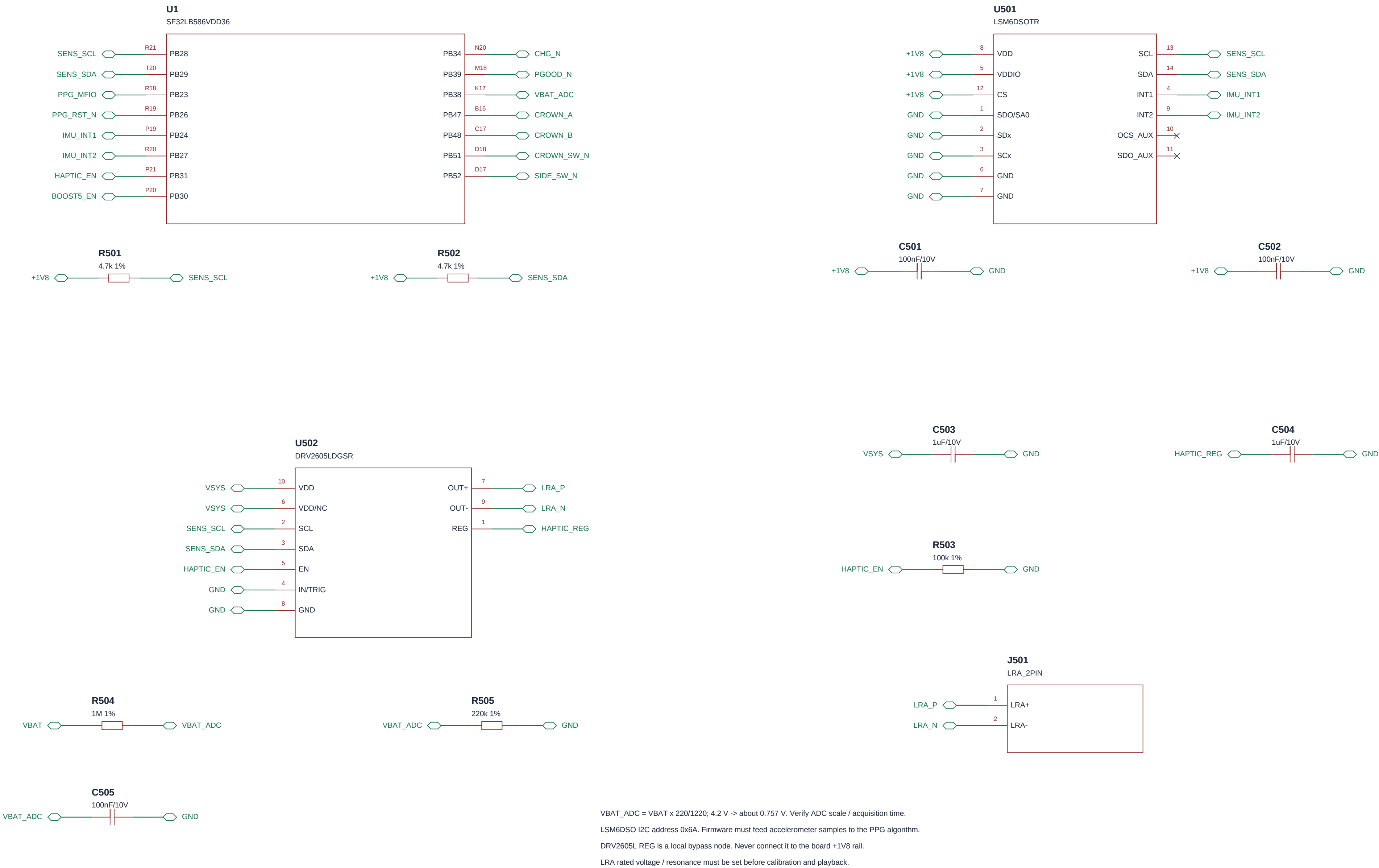


Encoder input uses firmware debounce; MCU ESD / mechanical switch package selection remains for PCB.

No software or display bandwidth claim substitutes for a measured animation frame rate on the purchased kit.

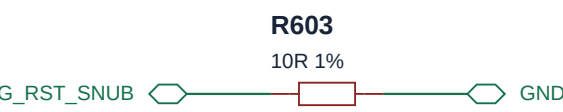
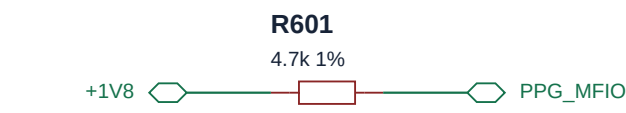
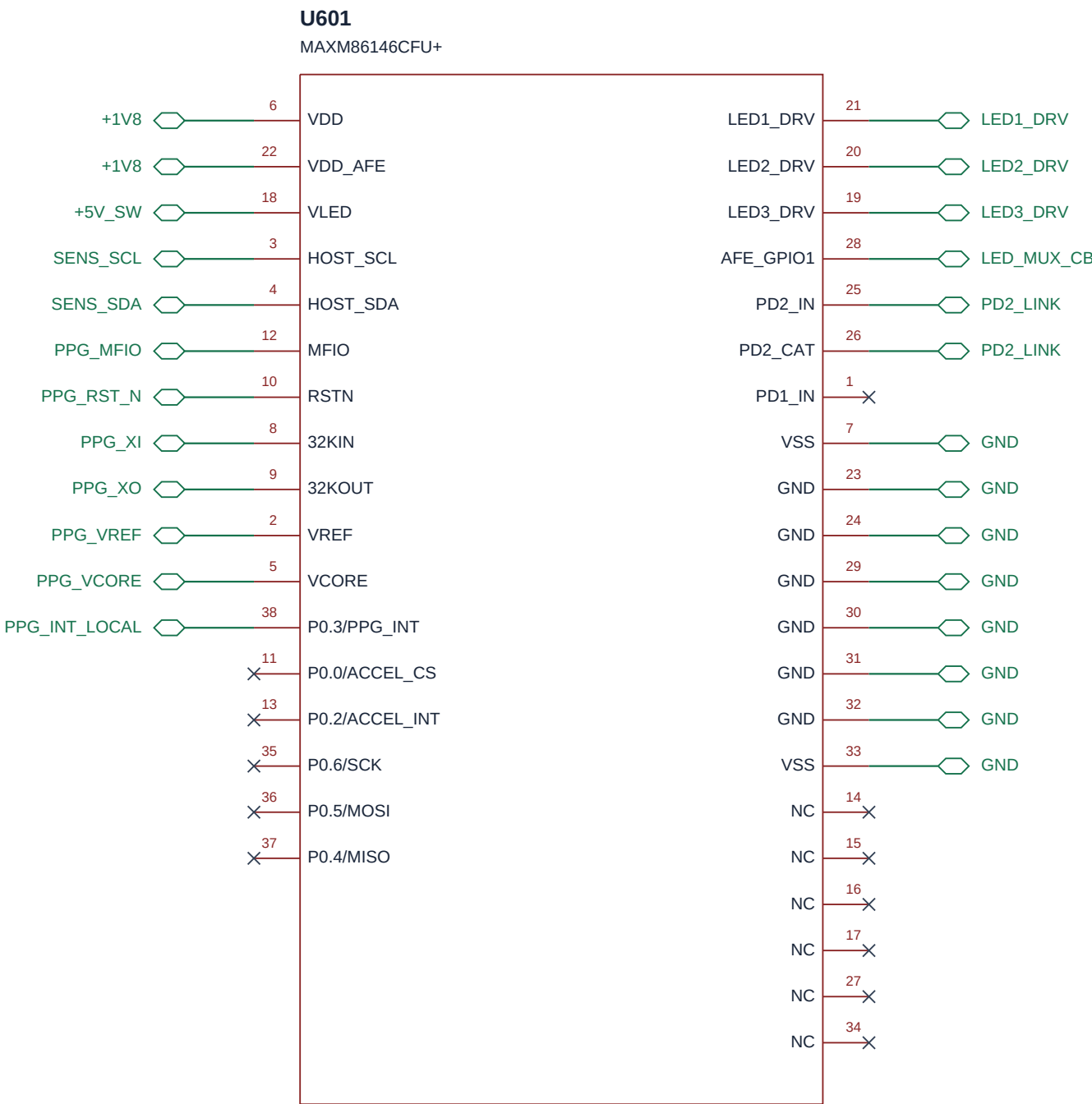
HealthWatch-SF58-BareChip-v0.3 | 08 IMU AND HAPTICS

LSM6DSO -> SF58 host -> MAXM86146 algorithm | 1.8 V sensor I2C bus | LRA with closed-loop driver



HealthWatch-SF58-BareChip-v0.3 | 09 PPG MODULE

MAXM86146CFU+ heart-rate / SpO2 front end | wrist optics and firmware algorithm still require validation



PD1 is integrated: pin 1 remains open. PD2 pin 25 must be joined to pin 26.

VCORE and VREF are capacitor-only nodes; do not drive with +1V8.

The unused accelerometer SPI pins stay unloaded in host-feed architecture.

PPG_INT_LOCAL is the local AFE interrupt; host communication uses MFIO.

Crystal load network follows the ADI reference; no guessed external load capacitors.

Host I2C / MFIO / reset all 1.8 V. I2C pull-ups are on sheet 08.

Hold PPG reset while required rails / clock stabilize, then follow ADI boot protocol.

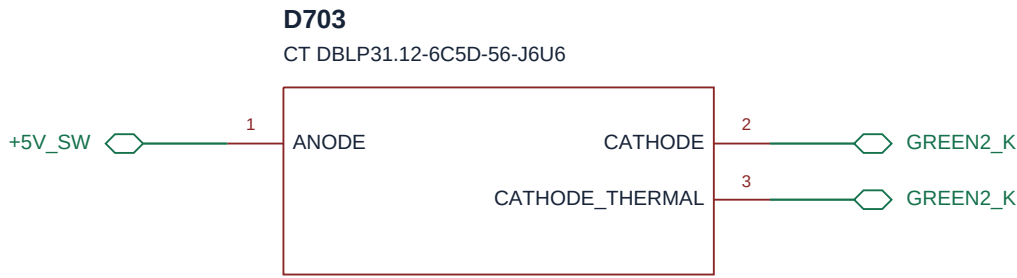
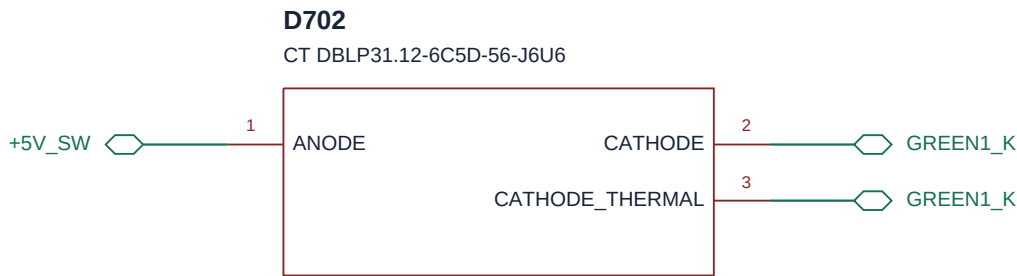
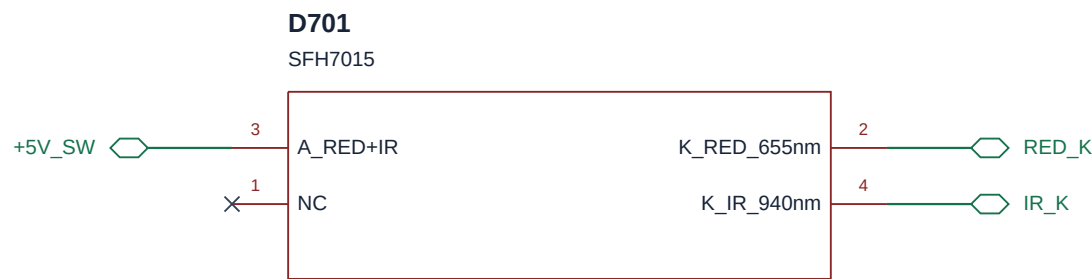
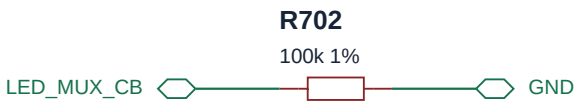
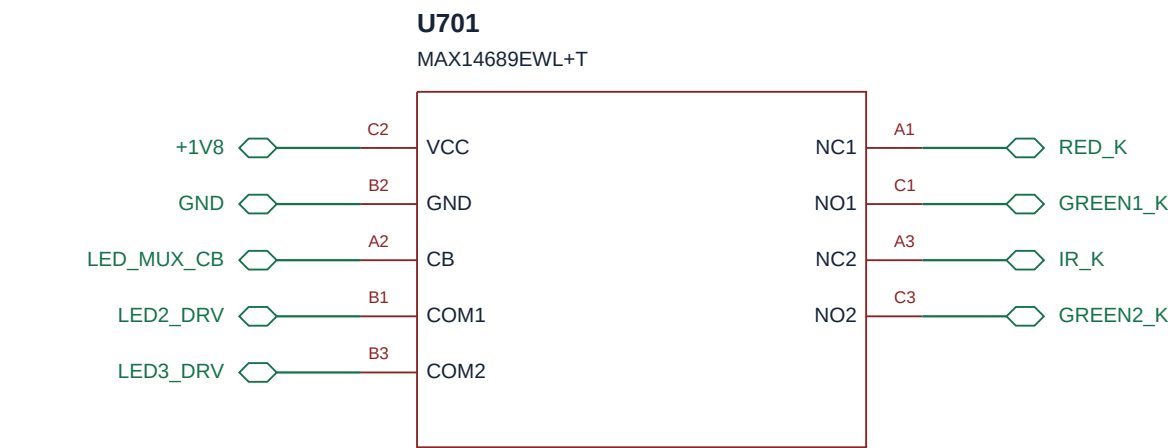
C601 at VDD; C602+C603 at VDD_AFE; C606+C607 at VLED.

Common ground electrically; keep LED pulse returns away from optical/analog nodes.

This is a real sensing circuit; accuracy is not established by the schematic.

HealthWatch-SF58-BareChip-v0.3 | 10 OPTICAL EMITTERS

Two green LEDs + red / IR | MAX14689 switching network | LED polarity / optical footprint review required



Current is regulated by MAXM86146 LED drivers. Set pulse current / timing before acquisition.

CB=0: LED2 -> RED, LED3 -> IR. CB=1: LED2 -> GREEN1, LED3 -> GREEN2.

GREEN1 cathode is also connected to LED1_DRV, following the ADI typical application.

Keep the GREEN1 shared path: three independent LEDs would change the reference design.

MAX14689 VCC=1.8 V; analog terminals support beyond-rail signals within its ratings.

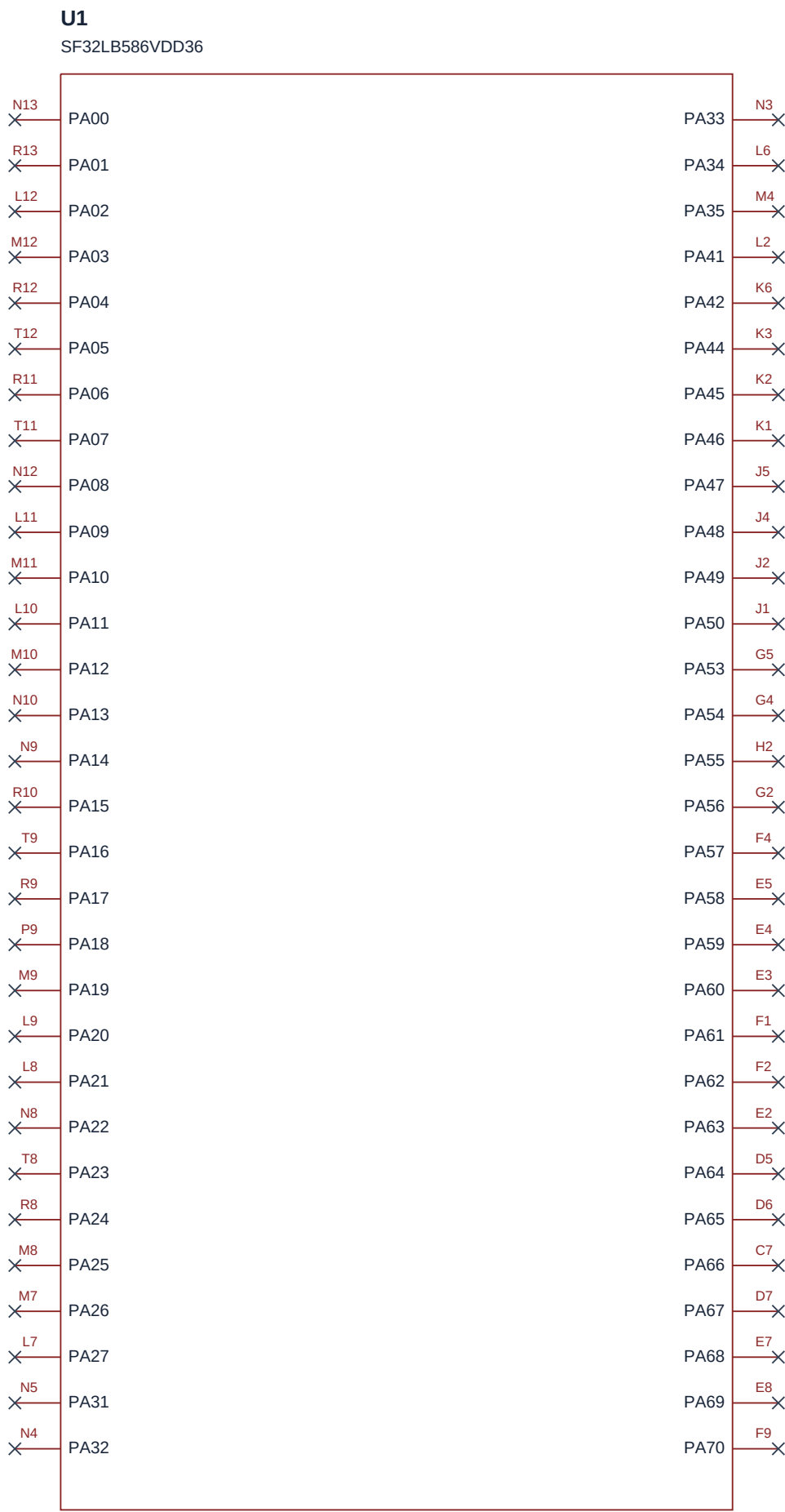
Green LED: physical pad 1 = anode; pads 2 and 3 = cathode. Both cathode pads are wired.

Before PCB: audit emitter land patterns, optical separation, black barriers and skin contact geometry.

Heart rate / SpO2 algorithms, calibration, motion rejection and human testing remain required.

HealthWatch-SF58-BareChip-v0.3 | 11 SPARE GPIO BALLS

Complete SF32LB586VDD36 pin inventory | spare pins not routed in this offline revision



U1 parts on every sheet are units of ONE physical BGA256 IC, with a shared unique ID and a single BOM entry.

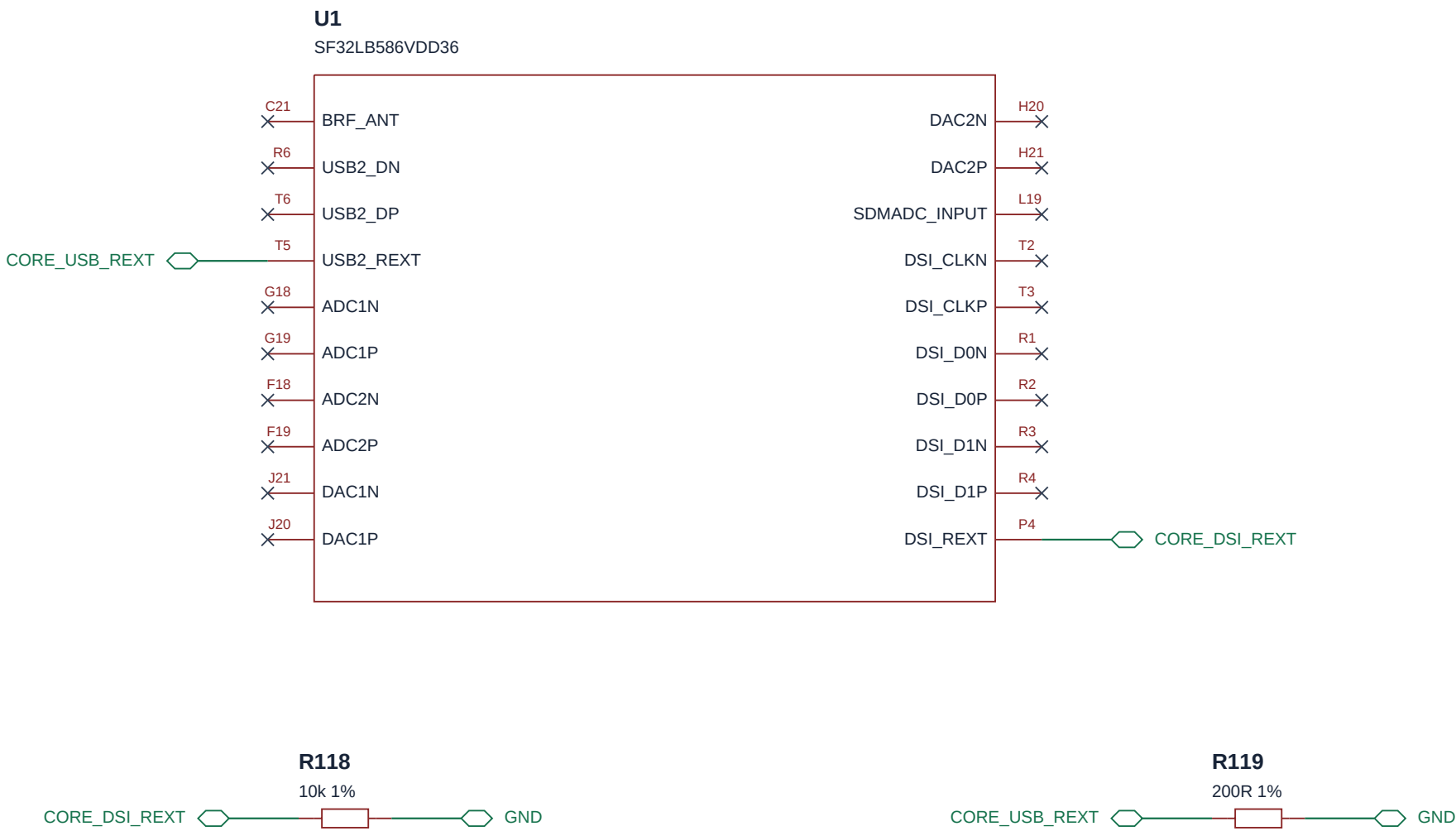
EDA no-connect flags mark deliberate unused pins. Configure unused digital inputs to avoid floating input current.

Ball names come from the official SiFii PADS symbol; full alternate-function names are in chip-ball-audit.csv.

Only actual device I/O is shown here. No module castellated-pad numbers remain in this project.

HealthWatch-SF58-BareChip-v0.3 | 12 GROUND AND UNUSED ANALOG

All BGA ground/reference-return balls | disabled RF, USB, audio and MIPI functions



All VSS/AVSS/PVSS balls are explicitly grounded, including A3/A4 listed as VSS in DS5801.

GPADC_VREFN, AUD_VREF_GND and SDMADC_VSS_VREF also return to ground; keep analog return paths quiet.

DSI_REXT 10k and USB2_REXT 200R follow the official circuit. Their buses are not connected in this offline revision.

BRF_ANT is deliberately open. Firmware must leave Bluetooth/RF disabled until an antenna network is designed.

B4/B5 external-core supply pins are open on BARE CHIP PMU, following the selected internal-PMU topology.

BGA pad geometry is taken from the official chip footprint. PCB placement/routing and fabrication release remain pending.

HealthWatch-SF58-BareChip-v0.3 | 13 TEST POINTS AND BRINGUP

1.0 mm probe pads | begin with a current-limited bench supply | no probes on crystal nodes



Power-up order: isolate external loads -> confirm VSYS / 1.8 V / 3.3 V -> reset and debug -> SiP / external Flash.

Then enable display and touch, IMU, LRA and PPG individually. Start PPG LED currents low and measure 5 V droop.

J101 is a 1.8 V debug interface. J202 / J402 / J501 are defined solder-wire pad banks, not purchased plug connectors.

Full system supply current includes charging and transient loads. The regulator current rating is not a rail budget.

A BOM/model pass or schematic DRC pass does not validate startup, sleep current, optics or PCB manufacturing.

Read the revision notes and bring-up checklist before ordering boards. No routed PCB or Gerber is included.